

NAVIN PALANIKUMAR

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Summary

AI-focused Backend Software Engineer with 2.5+ years of experience building scalable distributed systems, intelligent automation platforms, and cloud-native backend applications. Strong expertise in Java, Spring Boot, Kafka, PostgreSQL, Kubernetes, AWS, Azure, and AI-powered developer tooling using MCP and Claude. Experienced in designing high-throughput event-driven architectures, real-time streaming platforms, observability systems, and enterprise cloud infrastructure. Skilled in developing resilient microservices, optimizing large-scale data pipelines, and integrating **AI workflows into engineering productivity systems**. Proven track record in building Kafka streaming platforms processing 100K+ events/sec, improving API performance by 75%, automating operational workflows with AI agents, and implementing scalable infrastructure across Kubernetes and cloud environments. Passionate about Backend Engineering, AI Infrastructure, Cloud Platforms, Distributed Systems, Developer Productivity, and Intelligent Automation.

Skills

Programming languages: Java, Python, Ruby on Rails
Cloud infrastructure: AWS (Lambda, EC2, S3, EKS, CloudWatch, SNS, API Gateway, DynamoDB, RDS, IAM), Azure (AKS, Azure Service Bus, Azure Monitor, Log Analytics), Kubernetes, Docker, Helm, Terraform
DevOps and monitoring: Jenkins, ArgoCD, GitHub Actions, Airflow, Signoz, Prometheus, Grafana
Testing frameworks: JUnit, Mockito, Pytest

Backend frameworks: Spring Boot, Hibernate, JPA, REST APIs, OAuth, JMS, MCP (Model Context Protocol), AI agent integrations, Redis, Elasticsearch
Databases and messaging: PostgreSQL (PSQL), MySQL, DynamoDB, Snowflake, MongoDB, Redis, Apache Kafka (Kafka Streams, Kafka Connect), AKHQ schema registry
Architecture and engineering: Microservices architecture, event-driven architecture, distributed systems design

Experience

Software Engineer FourKites

06/2024 to Current
Chennai, IN

- Built scalable AI-enabled backend platforms using Java, Spring Boot, Kafka, PostgreSQL, Kubernetes, AWS, and Azure cloud services.
- Developed intelligent MCP (Model Context Protocol) server integrations with Claude Desktop to automate engineering workflows and developer productivity tasks.
- Designed high-throughput Kafka streaming systems processing 100K+ events/second with optimized partitioning, schema evolution, and consumer scaling strategies.
- Engineered resilient event-driven microservices with retry handling, dead-letter queues, circuit breakers, and fault-tolerant distributed communication.
- Improved API performance by 75% using advanced Redis caching, PostgreSQL query optimization, asynchronous processing, and connection pooling.
- Architected Kubernetes-based cloud-native deployments with horizontal pod auto-scaling, improving infrastructure scalability by 10x.
- Worked extensively with AWS services including Lambda, EKS, S3, DynamoDB, CloudWatch, SNS, and API Gateway for scalable backend infrastructure.
- Implemented Azure cloud solutions using AKS, Azure Monitor, Azure Service Bus, and Log Analytics for enterprise-scale workloads.
- Built observability and monitoring platforms using Signoz, Prometheus, Grafana, OpenTelemetry, and CloudWatch dashboards for proactive incident detection.
- Managed Kafka observability and cluster operations using AKHQ dashboarding and monitoring tools.
- Developed Airflow-based workflow orchestration pipelines for backend automation and large-scale data processing.
- Integrated Snowflake data pipelines and backend data workflows for scalable analytics and reporting systems.
- Executed zero-downtime Elasticsearch migration for 10TB+ production datasets using a double-write single-read migration architecture.

- Collaborated in architecture design reviews, production debugging, backend optimization, and platform reliability engineering.

Software Engineering Intern FourKites

07/2023 to 06/2024
Chennai, IN

- Developed backend services and REST APIs using Java, Spring Boot, and Ruby on Rails.
- Built foundational Kafka infrastructure including topic design patterns, schema integration, and consumer group management.
- Designed and optimized database schemas across PostgreSQL, DynamoDB, and MongoDB for high-volume data processing.
- Developed an intelligent alerting ChatBot that reduced operational overhead by 45% and minimized false-positive alerts.
- Created automated testing frameworks using JUnit, Mockito, Pytest, and TestContainers, improving code coverage and reducing manual testing effort.
- Implemented distributed tracing and monitoring solutions using Jaeger and custom metrics collection for performance analysis.
- Optimized CI/CD deployment pipelines with Jenkins and Docker, reducing deployment time from 45 minutes to 8 minutes.
- Developed robust ETL and automation pipelines processing over 1M+ records daily with retry and validation mechanisms.
- Automated repetitive operational workflows using Python scripts, significantly improving team productivity.
- Reduced production bugs by 30% through rigorous testing, debugging, and code quality improvements.

Education

Bachelor of Computer Science & Engineering Sri Krishna College Of Engineering And Technology

04/2024
Coimbatore

Key Technical Initiatives

- AI-Powered MCP Server with Claude Desktop – Built an MCP-based AI automation platform integrated with Claude Desktop to streamline engineering workflows, automate demo setup, and improve developer productivity using Java and Python.
- RCA MCP Agent – Developed an AI-powered Root Cause Analysis (RCA) agent to automate log analysis, issue investigation, and production debugging workflows for faster incident resolution.
- AI-Powered Jira Test Case Generator – Built an AI-driven Jira integration system to automatically generate and attach test cases from Jira tickets, reducing manual QA effort and improving testing efficiency.
- Zero-Downtime Elasticsearch Migration – Executed a large-scale Elasticsearch migration for 10TB+ production data using a zero-downtime double-write single-read strategy with reliable rollback and monitoring mechanisms.
- Enterprise Kafka Streaming Platform – Designed and implemented a high-throughput Kafka streaming platform processing 100K+ events/second with AKHQ observability, retry handling, and scalable event-driven architecture.
- ArgoCD Deployment Automation – Implemented ArgoCD-based GitOps deployment automation for Kubernetes applications to improve release reliability and streamline cloud-native CI/CD workflows.